

2017 F=ma Exam: Problem 22

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We are given

$$f_{\text{energy}} = \frac{\frac{1}{2}Mv_2^2}{\frac{1}{2}mv_0^2} = \frac{Mv_2^2}{mv_0^2}$$

where v_2 is the velocity of M after the collision.



For an elastic collision, we have

$$v_2 = \frac{2m}{m+M}v_0$$

Then

$$f_{\text{energy}} = \frac{M}{m} \left(\frac{2m}{m+M} \right)^2 = \frac{4Mm}{m^2 + 2mM + M^2} = \frac{4}{2 + r + \frac{1}{r}}$$

where $r = m/M$. To maximize f_{energy} , we want to minimize $r + \frac{1}{r}$ so we take $r = 1$. Thus, the answer is C.